KU LEUVEN DIFFERENCE-IN-DIFFERENCES WORKSHOP

Foundations of Difference-in-Differences

The 2×2 and Parallel Trends

Scott Cunningham · Baylor University

KU Leuven · Difference-in-Differences Workshop

About me

- **Scott Cunningham** — Professor of Economics, Baylor University; Visiting Professor of Government, Harvard University.
- This is a **one-day workshop on difference-in-differences**, in three parts, from the basic 2×2 through continuous treatment and staggered adoption.
- Much of the material is drawn from **Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (2026)**, "Difference-in-Differences: A Practitioner's Guide," *Journal of Economic Literature*, and new work on continuous-treatment DiD.

Three parts

Part	Topic
Part 1	2×2 DiD: parameters, calculations, and identification
Part 2	Continuous treatment DiD
Part 3	Staggered adoption DiD

Today: from the 2×2 to covariates

- What the 2×2 DiD is — as a **calculation** and as an **identification strategy**.
- Six equivalent ways to compute the 2×2 — and why that matters.
- What the DiD actually identifies: ATT under parallel trends.
- Event studies — what they tell you and what they don't.
- When covariates are needed and how to include them correctly.

Lesson

DiD is Two Things: Calculation + Assumption

DiD is a research design, not a regression

> *A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.*

1.		**2.**
A *calculation*	**+**	**An *assumption***
(the 2×2)		(parallel trends)

The individual treatment effect

$$\delta_i = Y_i(1) - Y_i(0)$$

- One per unit. **Different** across units in general — treatment effects are heterogeneous.
- We only ever see one of $Y_i(1)$ or $Y_i(0)$. The other is the counterfactual.
- So δ_i is never directly observed at the individual level.

The ATT: a causal parameter has three ingredients

$$\text{ATT} = \underbrace{\mathbb{E}\Big[}_{\text{weighted expectations}} \underbrace{Y(1) - Y(0)}_{\text{potential outcomes}} \underbrace{\Big| D = 1, \text{Post}\Big]}_{\text{population}}$$

- **Potential outcomes** — the contrast $Y(1) - Y(0)$
- **Population** — on whom: treated units, post-period
- **Weighted expectations** — how we average across that population

The switching equation: you see one half

$$Y_i \;=\; D_i \cdot Y_i(1) \;+\; (1 - D_i) \cdot Y_i(0)$$

- For each unit: $Y_i(1)$ if treated, $Y_i(0)$ if not.
- The ATT asks about *both* columns. The data hands you one.

The rest of today: strategies for recovering the ATT from the half we see.

LESSON

Six Equivalent Ways, One Number

Cheng & Hoekstra (2013): castle doctrine and homicide

The reform. “Castle doctrine” laws expanded the legal right to use lethal force in self-defense. Many US states passed expansions from **2005–2010**.

- **Panel:** all US states $\times$ years. **Outcome:** log homicide rate.
- **Treatment:** state passed a castle doctrine expansion.
- **Comparison:** states that did not pass (never-treated).

For today: a clean 2$\times$2 on **2005–2006**. Treated $=$ states adopting in 2006; control $=$ states that hadn’t. *Cheng & Hoekstra (2013), J. Human Resources 48(3): 821–854.*

The four cells, by hand on `castle.dta`

```
use castle.dta, clear
keep if year >= 2005 & year <= 2006
gen post = (year == 2006)
gen treat = (effyear == 2006)

summarize l_homicide if treat==1 & post==1 // Y11
summarize l_homicide if treat==1 & post==0 // Y10
summarize l_homicide if treat==0 & post==1 // Y01
summarize l_homicide if treat==0 & post==0 // Y00
```

$$\widehat{\text{DiD}} = (\overline{Y}_{11} - \overline{Y}_{10}) - (\overline{Y}_{01} - \overline{Y}_{00}) \;\approx\; \mathbf{0.10}$$

Log homicide rate $\sim 10\%$ *higher in 2006-adopting states relative to non-adopters.*

Four averages, three subtractions — two orderings

The 2×2 is four cell means: $\overline{Y}_{11}, \overline{Y}_{10}, \overline{Y}_{01}, \overline{Y}_{00}$.

First-difference ordering (→)

Subtract across time *within* each group, then compare groups:

$$\Delta_1 = \overline{Y}_{11} - \overline{Y}_{10}$$
$$\Delta_0 = \overline{Y}_{01} - \overline{Y}_{00}$$
$$\text{DiD} = \Delta_1 - \Delta_0$$

Group-difference ordering (↓)

Subtract across groups *within* each period, then compare periods:

$$G_{\text{post}} = \overline{Y}_{11} - \overline{Y}_{01}$$
$$G_{\text{pre}} = \overline{Y}_{10} - \overline{Y}_{00}$$
$$\text{DiD} = G_{\text{post}} - G_{\text{pre}}$$

Same four averages. Same three subtractions. Same number. The 2×2 is symmetric under reordering.

Six equivalent ways, one number

	Way	What it does	$\hat{\beta}_{DD}$
1.	First-difference arithmetic	$(\overline{Y}_{11} - \overline{Y}_{10}) - (\overline{Y}_{01} - \overline{Y}_{00})$	≈ 0.10
2.	Group-difference arithmetic	$(\overline{Y}_{11} - \overline{Y}_{01}) - (\overline{Y}_{10} - \overline{Y}_{00})$	≈ 0.10
3.	OLS with `post##treat`	saturated dummies, direct interaction	≈ 0.10
4.	Two-way fixed effects (TWFE)	absorbs unit $+$ time means	≈ 0.10
5.	ΔY_i on D_i (horizontal)	time-difference within group, then on D	≈ 0.10
6.	G_t on Post$_t$ (vertical)	group-difference within period, then on Post	≈ 0.10

Same data, six ways, identical to machine epsilon.

“DiD” is not a regression — there are six equivalent ways to compute it

- You will often hear: “I ran a diff-in-diff regression.”
- That is misleading. Two of the six ways are pure arithmetic; four are regressions. All give the same answer.
- The regressions are tools for requesting the estimate. The **2×2 is the thing being estimated**.

This is why we start with the 2×2, not the regression. The estimand is four averages and three subtractions. The regression is just a convenient way to compute — and standard-error — it.

LESSON

Decomposing the 2×2

The proof: add a zero, rearrange (four steps)

1. Start with the 2×2:

$$\widehat{\text{DiD}} = \big[\mathbb{E}[Y \mid 1, \text{Post}] - \mathbb{E}[Y \mid 1, \text{Pre}]\big] - \big[\mathbb{E}[Y \mid 0, \text{Post}] - \mathbb{E}[Y \mid 0, \text{Pre}]\big]$$

2. Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$:

$$\widehat{\text{DiD}} = \big[\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]\big] - \big[\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]\big]$$

3. Add a zero. Insert $+\,\mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}]$ (the missing counterfactual, twice):

$$\begin{aligned}\widehat{\text{DiD}} = {} & \mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}] \\ & + \mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}] - \big[\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]\big]\end{aligned}$$

4. Rearrange. First two terms $=$ ATT; remainder $=$ difference in untreated trends:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}\big[Y(1) - Y(0) \mid 1, \text{Post}\big]}_{\textbf{ATT}} + \underbrace{\Delta\mathbb{E}[Y(0) \mid 1] - \Delta\mathbb{E}[Y(0) \mid 0]}_{\text{PT bias } = \, 0 \text{ under parallel trends}}$$

The bias term *is* selection bias — on trends, not levels

$$\text{bias} = \underbrace{\Delta\mathbb{E}[Y(0) \mid D{=}1]}_{\text{how treated's } Y(0) \text{ moves}} - \underbrace{\Delta\mathbb{E}[Y(0) \mid D{=}0]}_{\text{how controls' } Y(0) \text{ moves}}$$

- **Cross-section selection bias**: are treated and control different in *levels* of $Y(0)$?
- **DiD's selection bias**: are treated and control different in *trends* of $Y(0)$?
- Different levels are fine — unit fixed effects absorb them.
- What can't differ is the *rate* at which $Y(0)$ moves across groups.

The trade DiD makes vs. randomization

RCTs guarantee identical trends (and levels) through the design itself.
DiD requires only one thing: parallel trends in $Y(0)$. But unlike randomization, **there is no guaranteed research design that delivers it** — it must be argued for on substantive grounds.

Ghanem, Sant'Anna & Wüthrich (2025).

Case 2: no-anticipation fails — treated group adjusts early

Treated units *anticipate* the policy and adjust before it lands. Their pre-period outcome is contaminated:

$$\mathbb{E}[Y \mid 1, \text{Pre}] \;=\; \mathbb{E}[Y(0) \mid 1, \text{Pre}] \;+\; \text{ATT}_{\text{pre}}$$

Apply the same $+0$ trick as Case 1. The contaminated pre-period subtracts off:

$$\widehat{\text{DiD}} \;=\; \underbrace{\text{ATT}_{\text{Post}}}_{\text{what we want}} \;+\; \underbrace{\Delta\mathbb{E}[Y(0) \mid 1] - \Delta\mathbb{E}[Y(0) \mid 0]}_{\text{PT bias} \,=\, 0} \;-\; \underbrace{\text{ATT}_{\text{pre}}}_{\text{anticipation bias}}$$

Numerical example

ATT $= 10$, PT bias $= 0$, $\text{ATT}_{\text{pre}} = 3$. $\widehat{\text{DiD}} = 10 - 3 = \mathbf{7}$. **Biased down.**

If effects are constant over time, $ATT_{pre} = ATT_{post}$ and DiD $\to 0$.

Case 3: SUTVA fails — control group is also treated in Post

The comparison group's Post outcome is $Y(1)$, not $Y(0)$ (spillover or simultaneous treatment):

$$\widehat{\text{DiD}} = \Big[\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]\Big] - \Big[\mathbb{E}[Y(1) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]\Big]$$

Add *two* zeros (one counterfactual on each side) and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}}_{\text{treated}} + \underbrace{\Delta\mathbb{E}[Y(0) \mid 1] - \Delta\mathbb{E}[Y(0) \mid 0]}_{\text{PT bias} = 0} - \underbrace{\text{ATT}_{D=0,\,\text{Post}}}_{\text{control's treatment effect}}$$

Numerical example

ATT $= 10$, PT bias $= 0$, $\text{ATT}_{D=0,\text{Post}} = 4$. $\widehat{\text{DiD}} = 10 - 4 = \mathbf{6}$. **Biased toward zero.**

Case 4: comparison group is always treated

The comparison group was treated *before* our data starts. Both their Pre and Post outcomes are $Y(1)$:

$$\widehat{\text{DiD}} = \Big[\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]\Big] - \Big[\mathbb{E}[Y(1) \mid 0, \text{Post}] - \mathbb{E}[Y(1) \mid 0, \text{Pre}]\Big]$$

Add $\pm\mathbb{E}[Y(0) \mid 0, \text{Post}]$ and $\pm\mathbb{E}[Y(0) \mid 0, \text{Pre}]$ on the comparison side and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}}_{\text{treated}} + \underbrace{\Delta\mathbb{E}[Y(0) \mid 1] - \Delta\mathbb{E}[Y(0) \mid 0]}_{\text{PT bias} = 0} - \underbrace{\Delta\,\text{ATT}_{D=0}}_{\text{change in comparison's TE}}$$

Numerical example

$\text{ATT} = 10$, PT bias $= 0$, $\Delta\text{ATT}_{D=0} = 2$. $\widehat{\text{DiD}} = 10 - 2 = 8$. **Biased toward zero.**

Four cases: one method

Case	What fails	$\widehat{\text{DiD}} =$
1. NA + untreated comparison	nothing (baseline)	ATT + PT bias
2. NA fails	treated Pre $\supset Y(1)$	ATT + PT bias − ATT_{pre}
3. Spillover (SUTVA)	control Post $\supset Y(1)$	ATT + PT bias − $\text{ATT}_{D=0,\text{Post}}$
4. Always-treated comparison	control Pre & Post $\supset Y(1)$	ATT + PT bias − $\Delta\text{ATT}_{D=0}$

Same method every time: write the 2×2, apply the switching equation, add the right zero(s), rearrange. Only what you add changes.

New Jersey raises its minimum wage. Pennsylvania doesn't.

- **April 1992**: NJ raises its minimum wage from \$4.25 to \$5.05 — a 19% increase.
- Pennsylvania, just over the river, stays at \$4.25.
- Card and Krueger survey 410 fast-food restaurants on both sides of the border, before (Feb 1992) and after (Nov 1992).
- The natural 2×2: **treated = NJ, control = PA, pre/post the policy.**

Card & Krueger, American Economic Review, 1994.

The 2×2 as long differences: horizontal regression

Step 1: Within each group, subtract across time (pre $\to$ post).

	Pre	Post	Δ ($\to$)
NJ (treated)	20.44	21.03	$+0.59$
PA (control)	23.33	21.17	-2.16

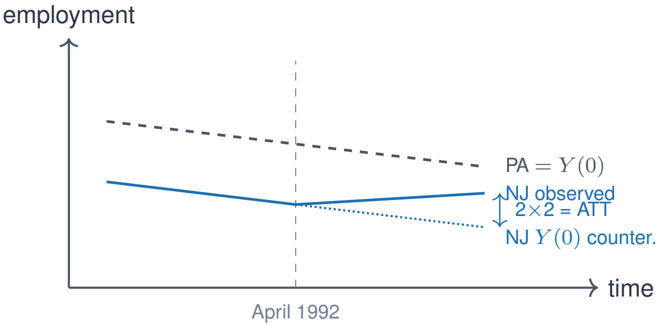
Step 2: Subtract the two time changes:

$$\widehat{\beta}_{\text{DiD}} = (+0.59) - (-2.16) = \mathbf{+2.75}$$

Imbens: the **long-difference regression** (or **horizontal regression**)

$\Delta Y_i = \alpha + \beta_{\text{DiD}} D_i + u_i$ — difference over time *within* each group, then compare groups.

Parallel trends visualized: the control *is* the counterfactual



The dotted line is what NJ would have done absent the policy. We never see it.

PT lets PA's slope stand in for it.

The same 2×2 as group differences: vertical regression

Step 1: At each time period, subtract across groups (NJ−PA, ↓).

	Pre	Post
NJ (treated)	20.44	21.03
PA (control)	23.33	21.17
Gap (NJ−PA ↓)	-2.89	-0.14

Step 2: Subtract the two group gaps:

$$\widehat{\beta}_{\text{DiD}} = (-0.14) - (-2.89) = \textbf{+2.75}$$

Imbens: the **group-differences regression** (or **vertical regression**)

$D_t = \alpha + \beta_{\text{DiD}}\,\text{Post}_t + v_t$ — compare groups *within* each period, then difference across time.

Two orderings, one number — and why the vertical view matters

Ordering	Regression	Estimate
Time first, groups second ($\rightarrow$ then $\uparrow$)	Long-difference (horizontal)	$+2.75$
Groups first, time second ($\downarrow$ then $\rightarrow$)	Group-differences (vertical)	$+2.75$

- Both orderings of the double-difference give the same number.
- The 2$\times$2 is symmetric under reordering — it doesn't matter which subtraction comes first.

Why the vertical view matters (Imbens 2021): Allow an intercept in the group-differences regression: $D_t = \mu + \beta_{\text{DiD}}\,\text{Post}_t$. That μ *absorbs the level difference between the two groups* — which is the key move in synthetic control. No convex-hull constraint required.

Three flavors of standard error

- **Vanilla (homoskedastic)**: assumes $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2$ for all i.
- **Robust (HC)**: each residual speaks for itself — handles heteroskedasticity.
- **Cluster-robust (CRVE)**: groups residuals by cluster; within-cluster errors correlate freely.

Same point estimate, three different SEs. The SE you pick is a claim about the error structure.

Bertrand, Duflo & Mullainathan (QJE 2004): the placebo-law test

"How Much Should We Trust Differences-in-Differences Estimates?"

- BDM generate *placebo laws*: assign random treatment dates to states in a real CPS panel.
- True effect by construction: $\beta = 0$. So $\Pr(\text{reject } H_0)$ should equal $\alpha = 5\%$.
- **Vanilla SE**: rejection rate $\approx \mathbf{45\%}$. Wildly over-rejects.
- **HC-robust**: rejection rate *also* $\approx 45\%$. **HC barely moves the needle.**
- **Cluster-robust at state level**: rejection rate $\approx \mathbf{5\%}$. The fix.

Why HC doesn't save you

Binary, balanced D + within-state serial correlation. The het-correction has nothing to do with the actual problem. **Clustering is doing all the work.**

LESSON

Five Pieces of a Strong DiD

Five pieces of a strong DiD

1. **Bite** — the policy actually moved what it was supposed to move.
2. **Event study** — pre/post dynamics around the treatment date.
3. **Falsification** — a rival theory fails on a placebo group or outcome.
4. **Main result** — the DiD on the outcome that matters.
5. **Mechanism** — the causal pathway from D to Y.

We will walk through all five using a single case study: Medicaid expansion and near-elderly mortality.

From 2×2 to 2×T: the event study regression

Extend from one pre/post comparison to a full timeline:

$$Y_{it} \;=\; \alpha_i \;+\; \tau_t \;+\; \sum_{k \neq -1} \beta_k \cdot \big(D_i \cdot \mathbf{1}\{t = k\}\big) \;+\; \varepsilon_{it}$$

- α_i, τ_t — unit and time fixed effects. $t = -1$ is omitted (the reference period).
- Each $\hat{\beta}_k$ is a 2×2 DiD against $t = -1$: $\hat{\beta}_k = \big(\bar{Y}^T_k - \bar{Y}^T_{-1}\big) - \big(\bar{Y}^C_k - \bar{Y}^C_{-1}\big)$.
- Pre-period ($k < -1$): placebo estimates — should be near zero under PT.
- Post-period ($k \geq 0$): dynamic treatment effects.

The event study gives evidence, not identification

The 2×2 estimator recovers:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}}_{\text{target}} + \underbrace{\Big(\mathbb{E}[\Delta Y(0) \mid D{=}1] - \mathbb{E}[\Delta Y(0) \mid D{=}0]\Big)}_{\text{PT bias}}$$

- **PT bias** $=$ the gap between treated and control *untreated* trends.
- Parallel trends *assumes* this is zero — we never observe $Y(0)$ for the treated post-treatment.
- Pre-period placebo coefficients ($\hat{\beta}_k$, $k < -1$) ask: was this bias near zero *before*?

An event study is **not** needed for identification — PT is an assumption, not a testable claim.

It is needed for **evidence**: flat pre-trends suggest the bias was small in the past, which makes PT plausible going forward.

Medicaid expansion and near-elderly mortality

Miller, Johnson & Wherry (2021), *Quarterly Journal of Economics*.

- Did the ACA Medicaid expansion (2014) reduce mortality among the **near-elderly** (ages 55–64)?
- Comparison: states that expanded Medicaid vs. states that didn't, before vs. after 2014.
- Headline finding: $\approx 9.2\%$ **reduction** in near-elderly mortality.

Why *near*-elderly?

Everyone 65+ already has Medicare — Medicaid expansion is irrelevant for them. The elderly become a natural **placebo group** living in the same states and the same macroeconomy.

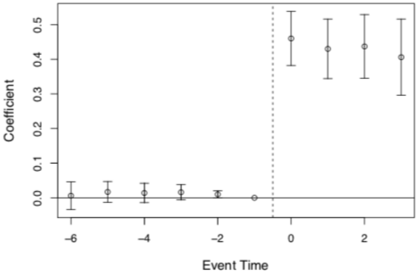
Strategy: walk through all five pieces of a strong DiD, in order.

Five pieces — start with bite

1. **Bite** — the policy actually moved what it was supposed to move.
2. *Event study*
3. *Falsification*
4. *Main result*
5. *Mechanism*

If the policy didn't move the treatment variable, nothing downstream is credible.

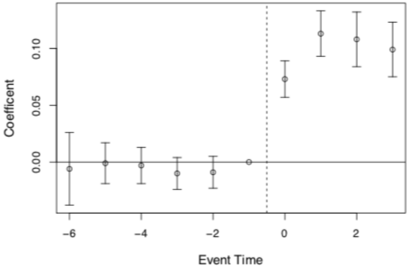
Bite #1: Medicaid take-up jumped in expansion states



(a) Medicaid Eligibility

Did the policy move people onto Medicaid? *Yes — enrollment rose sharply in expansion states post-2014, flat in non-expansion states. If the law didn't actually do something, nothing downstream is attributable to it.*

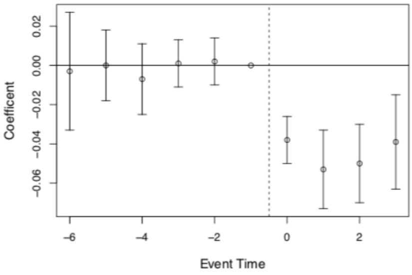
Bite #2: any-insurance coverage also rose



(b) Medicaid Coverage

Did Medicaid crowd out private insurance, or did total coverage rise? *Coverage rose broadly in expansion states — take-up is real, not mere substitution from one insurer to another.*

Bite #3: the uninsured rate fell in expansion states



(c) Uninsured

Third measure, same story. *Uninsured rate dropped sharply in expansion states post-2014 and stayed flat in non-expansion states. Three independent measures of bite all point the same direction.*

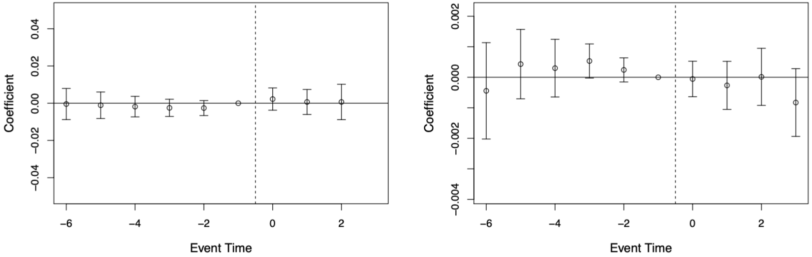
Five pieces — now falsify

1. *Bite* ✓
2. *Event study*
3. **Falsification** — a rival theory fails on a placebo group or outcome.
4. *Main result*
5. *Mechanism*

A rival explanation: some 2014 shock hit expansion states. If true, it would also hit the 65+ population. It didn't.

Falsification: same model, 65+ population — no effect

Age 65+ in 2014



(c) Medicaid Coverage (d) Annual Mortality

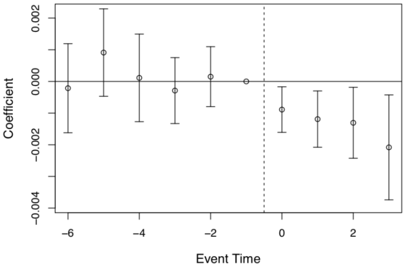
Same states, same years, same model — but run on the 65+ population. *They all have Medicare, so Medicaid expansion shouldn't touch their mortality. A rival story ("some 2014 shock hit expansion states") would predict a mortality effect here. It didn't appear.* ***Rival rejected.***

Five pieces — event study and main result

1. *Bite* ✓
2. **Event study** — pre/post dynamics around the treatment date.
3. *Falsification* ✓
4. **Main result** — the DiD on the outcome that matters.
5. *Mechanism*

Flat pre-trends build credibility. The post-2014 break is the result.

Main result: $\approx 9.2\%$ reduction in near-elderly mortality



- Post-2014: mortality declines in expansion states. $\approx 9.2\%$ **reduction.**
- Pre-period $\hat{\beta}_k$ near zero. Wide CIs — but the event study + falsification + bite tell the full story.
- A critic must now explain bite *and* falsification *and* the dynamic post-trend simultaneously.

Five pieces — mechanism

1. *Bite* ✓
2. *Event study* ✓
3. *Falsification* ✓
4. *Main result* ✓
5. **Mechanism** — the causal pathway from D to Y.

What kind of mortality falls? Disease-specific decomposition tells us where Medicaid is operating.

Mechanism: disease-related deaths fall

- Mechanisms are the part of a result that is **transportable** to other settings and populations.
- Reduced-form results often bundle several mechanisms — decompositions separate them.
- Here: **disease-related deaths fall** (cardiovascular, respiratory, infectious disease); injury deaths do not.
- Consistent with **chronic-disease management** — exactly what insurance-mediated access to care would facilitate.

Mechanisms tell you *why* it worked and *how far the result generalizes*. If the mechanism is access to preventive care, the finding transfers to other populations with similar unmet preventive needs.

The five pieces, all five present in Medicaid

Piece	Test	Result
Bite	Enrollment, any-insurance, uninsured rate	**All three move**
Event study	Pre-$\hat{\beta}_k$ near zero; 2014 break	**Flat pre-trends**
Falsification	Same model, 65$+$ population (Medicare)	**No effect**
Main result	DiD on near-elderly mortality	-9.2%
Mechanism	Disease-specific decomposition	**Chronic disease**

A critic must explain all five simultaneously. The evidentiary burden is now very high.

LESSON

Covariates in DiD

Recall: the 2×2 is DiD — six equivalent forms

1. $\underbrace{\big[\mathbb{E}[Y|D{=}1,t]-\mathbb{E}[Y|D{=}1,t{-}1]\big]}_{\text{first difference}}-\underbrace{\big[\mathbb{E}[Y|D{=}0,t]-\mathbb{E}[Y|D{=}0,t{-}1]\big]}_{\text{first difference}}$
2. $\underbrace{\big[\mathbb{E}[Y|D{=}1,t]-\mathbb{E}[Y|D{=}0,t]\big]}_{\text{group difference}}-\underbrace{\big[\mathbb{E}[Y|D{=}1,t{-}1]-\mathbb{E}[Y|D{=}0,t{-}1]\big]}_{\text{group difference}}$
3. $Y=\alpha_1+\alpha_2\,\mathrm{Post}+\alpha_3\,D+\delta\,(\mathrm{Post}\times D)+\varepsilon$ *saturated regression*
4. $Y=\alpha_1+\tau_t+\sigma_i+\delta\,(\mathrm{Post}\times D)+\varepsilon$ *two-way fixed effects*
5. $\Delta Y_i=\alpha_1+\delta\,D_i+\varepsilon_i$ *first-difference / horizontal regression*
6. $\Delta\bar{Y}_G=\alpha_1+\delta\,\mathrm{Post}_G+\varepsilon_G$ *group / vertical regression*

All six are **numerically identical**.

The 2×2 DiD estimand

$$\underbrace{\widehat{\delta}_{\text{DiD}}}_{\text{what we estimate}} = \underbrace{\text{ATT}}_{\text{what we want}} + \underbrace{\Big[\mathbb{E}[\Delta Y(0) \mid D{=}1] - \mathbb{E}[\Delta Y(0) \mid D{=}0]\Big]}_{\text{PT bias}}$$

Treatment = college education. Outcome = wages.
Parallel trends requires the PT bias term to equal zero.

A thought experiment

Group	Untreated wage trend $\Delta Y(0)$
Male workers	$+10$ per year
Female workers	$+8$ per year

Case 1: $D{=}1$ is 75% male. $D{=}0$ is 75% male.

Will parallel trends hold?

Case 1: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D{=}1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D{=}0]$	$0.75 \times 10 + 0.25 \times 8$	9.5

Parallel trends holds? **Yes.**

$\widehat{\delta}_{\text{DiD}} = \text{ATT}$ ✓

Case 2: unequal composition

	% male
$D=1$ (college educated)	75%
$D=0$ (no college)	25%

Same wage trends as before ($+10$ male, $+8$ female).
Will parallel trends hold now?

Case 2: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D{=}1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D{=}0]$	$0.25 \times 10 + 0.75 \times 8$	8.5

Parallel trends holds? **No.**

$\widehat{\delta}_{\text{DiD}} \neq$ ATT because PT bias $= 9.5 - 8.5 = 1.0 \neq 0$

Why? Imbalance on a trend-causing covariate

DiD does **not** require the two groups to look the same on *levels*.

DiD **does** require the same $Y(0)$ *trends*.

Imbalance in a covariate that *causes* $Y(0)$ trends
will **mechanically** break parallel trends.

Fix: control for sex

If sex is the only covariate driving differential trends,
then *conditioning on sex* removes the PT bias.

Several ways to do this:

1. TWFE saturated regressions
2. Inverse probability weighting (IPW)
3. Outcome regression (OR)
4. Doubly robust (DR)

We will discuss all four.

For conditioning to work: conditional PT

We need parallel trends *within* each group:

1. Male treated workers have the *same* $Y(0)$ trend as male control workers
2. Female treated workers have the *same* $Y(0)$ trend as female control workers

Conditional PTA: $\mathbb{E}[\Delta Y(0) \mid D{=}1,\ X] = \mathbb{E}[\Delta Y(0) \mid D{=}0,\ X]$ for all X

The standard TWFE with additive X

$$Y_{it} \;=\; \alpha_i + \lambda_t + \delta\,(D_i \times \mathrm{Post}_t) + \boldsymbol{\beta}\,X_i + \varepsilon_{it}$$

Two hidden assumptions:

1. Returns to X are **constant over time**
2. Treatment effect is **the same for all X**

Fatal problem: if X_i is time-invariant, it is collinear with α_i and drops out — whether you use unit fixed effects *or* a saturated dummy specification. An additive, separable X cannot deliver conditional parallel trends if the covariate is absorbed before it can do any work.

What if returns to X change over time?

Example: discrimination against women is not constant — $\beta_{\text{post}} \neq \beta_{\text{pre}}$.

Allow time-varying returns in the DGP:

$$Y_{it}(0) = \alpha_i + \gamma_t + \beta_t \, X_i$$

Write out the **post** and **pre** equations for unit i:

$$Y_{i,\text{post}}(0) = \alpha_i + \gamma_{\text{post}} + \beta_{\text{post}} \, X_i$$

$$Y_{i,\text{pre}}(0) = \alpha_i + \gamma_{\text{pre}} + \beta_{\text{pre}} \, X_i$$

Subtract (post $-$ pre):

$$\Delta Y_i(0) = \underbrace{(\gamma_{\text{post}} - \gamma_{\text{pre}})}_{\Delta\gamma} + \underbrace{(\beta_{\text{post}} - \beta_{\text{pre}})}_{\Delta\beta} \cdot X_i$$

Now take expectations by group

Starting from $\Delta Y_i(0) = \Delta\gamma + \Delta\beta \cdot X_i$, condition on treatment status:

$$E[\Delta Y(0) \mid D{=}1] = \Delta\gamma + \Delta\beta \cdot E[X \mid D{=}1]$$
$$E[\Delta Y(0) \mid D{=}0] = \Delta\gamma + \Delta\beta \cdot E[X \mid D{=}0]$$

Parallel trends bias:

$$\underbrace{E[\Delta Y(0) \mid D{=}1] - E[\Delta Y(0) \mid D{=}0]}_{\text{PT bias}} \;=\; \Delta\beta \;\cdot\; \big(\, E[X \mid D{=}1] - E[X \mid D{=}0] \,\big)$$

PT fails whenever **both**: returns shift ($\Delta\beta \neq 0$) **and** groups are imbalanced on X.

Fix for Bias #1: interact $\mathrm{Post} \times X$

$$Y_{it} \;=\; \alpha_i + \lambda_t + \delta\,(D_i \times \mathrm{Post}_t) + \beta\,X_i + \boldsymbol{\gamma}\,(\mathrm{Post}_t \times X_i) + \varepsilon_{it}$$

γ absorbs any shift in returns to X between pre and post (Spec B in the lab).

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.agecube ///
    i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##i.u74, robust
```

LESSON

Bias #2: Heterogeneous Treatment Effects

What if the effect of college differs by sex?

Previously we worried about **trends in** $Y(0)$ being unequal across groups.

Now the concern is different:

$$\mathbb{E}[Y(1) - Y(0) \mid \text{Male}] \neq \mathbb{E}[Y(1) - Y(0) \mid \text{Female}]$$

Do the rich and the poor get the same return to college?
Do workers in cities and towns get the same return?

If not, **and** the treatment group is imbalanced on these covariates,
the 2×2 DiD will not recover the ATT.

Proof, step 1 — the TWFE model and conditional PT

The TWFE model with one covariate X:

$$Y_{it} \;=\; \alpha_i + \gamma_t + \delta\,(T_t \times D_i) + \beta\,X_i + \varepsilon_{it}$$

Conditional parallel trends says, for each value of X:

$$\mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D{=}1,\; X] \;=\; \mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D{=}0,\; X]$$

Rearranged — the missing counterfactual for the treated is identified:

$$\mathbb{E}[Y(0)_{\text{post}} \mid D{=}1, X] \;=\; \mathbb{E}[Y_{\text{pre}} \mid D{=}1, X] + \mathbb{E}[Y_{\text{post}} \mid D{=}0, X] - \mathbb{E}[Y_{\text{pre}} \mid D{=}0, X]$$

Proof, step 2 — plug TWFE in: the slopes look the same

From the TWFE model, observed conditional means:

$$\mathbb{E}[Y_{\text{pre}} \mid D{=}1, X] = \alpha_1 + \alpha_3 + \beta\, X$$
$$\mathbb{E}[Y_{\text{post}} \mid D{=}0, X] = \alpha_1 + \alpha_2 + \beta\, X$$
$$\mathbb{E}[Y_{\text{pre}} \mid D{=}0, X] = \alpha_1 + \beta\, X$$

Counterfactual for treated (step 1 identity):

$$\mathbb{E}[Y(0)_{\text{post}} \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \beta\, X$$

Treated outcome, post: $\mathbb{E}[Y(1)_{\text{post}} \mid D{=}1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta\, X$

Both potential outcomes carry the *same* β. Subtract: ATT $= \delta$. *Suspiciously clean.*

Proof, step 3 — what if $Y(1)$ and $Y(0)$ have different slopes in X?

Let the slopes differ: write β_1 for the $Y(1)$ slope, β_2 for $Y(0)$:

$$\mathbb{E}[Y(1)_{\text{post}} \mid D{=}1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta_1 X$$
$$\mathbb{E}[Y(0)_{\text{post}} \mid D{=}1, X] = \alpha_1 + \alpha_2 + \alpha_3 \qquad + \beta_2 X$$

True ATT given X:

$$\text{ATT}(X) \ = \ \delta + (\beta_1 - \beta_2)\, X$$

- If $\beta_1 \neq \beta_2$, the treatment effect *depends on* X
- TWFE fits one β, not two — so $\hat{\delta}$ recovers ATT only if $\beta_1 = \beta_2$

Proof, step 4 — TWFE imposes homogeneous δ in X

Assumption: TWFE-with-X requires the treatment effect to be the same at every value of X.

- If X is **sex**: effect must be equal for men and women
- If X is **income**: effect must be equal at $1 and at $1,000,000
- If X is **age**: effect must be equal at 25 and at 65

OLS doesn't warn you that you're making this assumption. The math does.
If effects actually vary with X, TWFE averages across the variation
and returns a weighted parameter you didn't ask for.

Fix for Bias #2: saturate treatment with X

1. **Run a saturated regression on first differences** — interact treatment with every X
2. **Predict $\Delta Y(1)$ and $\Delta Y(0)$ for each unit** — average the gap in the treated group

```
* 1. Saturated regression on first differences
reg dy i.ever_treated ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.educsq ///
    i.ever_treated#i.marr i.ever_treated#i.nodegree ///
    i.ever_treated#i.black i.ever_treated#i.hisp ///
    i.ever_treated#c.re74 i.ever_treated#i.u74 ///
    $X, robust

* 2. Predict potential outcomes; average the gap in the treated group
replace ever_treated = 1
quietly predict dy_hat_1, xb
replace ever_treated = 0
quietly predict dy_hat_0, xb
replace ever_treated = evt_orig // restore
gen tau_sat = dy_hat_1 - dy_hat_0
quietly sum tau_sat if ever_treated == 1 // ATT = r(mean)
```

Bias #3: time-invariant covariates get absorbed

If you need X_i for conditional PT, but X_i doesn't change over time,
TWFE will **delete it**.

TWFE "demeans" the data. The mean of a constant is itself.
So $X_i - \bar{X}_i \equiv 0$ — it drops out of the regression entirely.

Fix: include $X_i \times \mathrm{Post}_t$ instead. The Post interaction is time-varying,
so it survives demeaning and can do the work of conditioning on X.

Three biases — one family of fixes

Bias	What TWFE imposes	Fix
#1 Trend imbalance	β constant pre and post	$\mathrm{Post} \times X$
#2 Heterogeneous TE	δ same at every X	$D \times X$ (saturate treatment)
#3 Time-invariant X	X_i survives demeaning	$\mathrm{Post} \times X$ (same fix as #1)

Most researchers include additive X and call it a day. That is the wrong spec. To fix all three you need both $\mathrm{Post} \times X$ and $D \times X$. It is rare to see this in published work.

Now: the LaLonde application

Switch to **02b-lalonde.pdf**

We will walk through the code together,
then return here for OR, IPW, and DR.

Lesson

Alternative Estimators

Three more estimators

Now we will look at three more estimators.

These are alternatives to the regression-based specs we just covered.

1. **IPW** — reweight controls to look like treated
2. **Outcome regression (OR)** — model the control trend, impute the counterfactual
3. **Doubly robust (DR)** — combine both; consistent if *either* is right

IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^{n} \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \, \Delta Y_i$$

$\hat{p}(X_i) = \widehat{\Pr}(D_i = 1 \mid X_i)$ (propensity score) $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$

IPW: five steps

1. **Estimate $\hat{p}(X)$.** Logit of D_i on X_i.
2. **Check overlap.** Plot $\hat{p}$ by treatment status. Trim if needed.
3. **Construct weights.** $w_i = (D_i - \hat{p}(X_i))/(1 - \hat{p}(X_i))$.
4. **Weighted average.** $\widehat{\mathrm{ATT}} = \sum w_i \, \Delta Y_i \; / \; n_T$.
5. **Standard error.** Bootstrap or influence function.

Overlap is non-negotiable

If $\hat{p}(X_i) \to 1$ for a control unit, its weight $\to \infty$.

No overlap = no valid counterfactual.

This is a data problem. No estimator fixes it.

OR — or “regression adjustment”

$$\widehat{\mathrm{ATT}}_{\mathrm{OR}} = \frac{1}{n_T} \sum_{D_i=1} \Big[\Delta Y_i - \hat{\mu}_0(X_i) \Big]$$

$\hat{\mu}_0(X_i) = \widehat{E}[\Delta Y \mid D{=}0,\ X]$ estimated on controls only

The estimator is increasingly called **regression adjustment** (RA) in the modern applied literature. Same idea, newer name.

OR: five steps

1. **First-difference.** $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$.
2. **Fit outcome model.** Regress ΔY on X *using controls only*.
3. **Predict counterfactual.** Evaluate $\hat{\mu}_0(X_i)$ for each treated unit.
4. **Mean gap.** ATT $= \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$.
5. **Standard error.** Bootstrap.

Surprise — you already know OR

OR step 2: *"Regress ΔY on X using controls only."*

That is exactly what the saturated TWFE does.

Let's prove it slowly.

Proof, step 1 — the saturated model

Spec C runs this regression on first differences:

$$\Delta Y_i \;=\; \underbrace{\alpha}_{\text{intercept}} \;+\; \underbrace{\delta\, D_i}_{\text{treatment}} \;+\; \underbrace{\beta'\, X_i}_{\text{covariates}} \;+\; \underbrace{\tau'\,(D_i \times X_i)}_{\text{treatment interactions}} \;+\; \varepsilon_i$$

$D_i \in \{0,1\}$ and **every** element of X_i is interacted with D_i.

Proof, step 2 — separate by group

Plug $D_i = 0$ and $D_i = 1$ into the model:

$$D_i = 0: \quad \Delta Y_i = \alpha + \beta' X_i + \varepsilon_i$$

$$D_i = 1: \quad \Delta Y_i = (\alpha + \delta) + (\beta + \tau)' X_i + \varepsilon_i$$

The two groups have **entirely different** parameters.

Controls contribute to (α, β). Treated contribute to $(\alpha + \delta, \beta + \tau)$.

Proof, step 3 — OLS separates completely

Re-label: $a_0 = \alpha,\ b_0 = \beta,\ a_1 = \alpha + \delta,\ b_1 = \beta + \tau.$

The OLS objective is:

$$\min_{a_0,b_0,a_1,b_1} \underbrace{\sum_{D_i=0} (\Delta Y_i - a_0 - b_0'\,X_i)^2}_{\text{only controls}} + \underbrace{\sum_{D_i=1} (\Delta Y_i - a_1 - b_1'\,X_i)^2}_{\text{only treated}}$$

The two sums share **no parameters**.

$(\hat{a}_0,\, \hat{b}_0)$ = OLS of ΔY on X **using controls only.**

Proof, step 4 — Spec C and OR make the same imputation

For a treated unit i, what does each estimator predict as $\Delta Y_i(0)$?

Spec C (saturated TWFE)	set $D_i{=}0$:	$\hat{a}_0 + \hat{b}_0' \, X_i$
OR / **regression adjustment**	control OLS:	$\hat{a}_0 + \hat{b}_0' \, X_i$

Identical prediction $\Rightarrow$ identical ATT. **Spec C is OR.**

One estimator, two names — just like the 2×2

Recall: six formulas for the 2×2, all giving the same number.

Same principle here.

Saturated TWFE (Spec C) and outcome regression (OR / RA)
are **the same estimator** expressed two different ways.

We will confirm this in the do file: both print $1,770.

DR: two chances to be right

$$\widehat{\mathrm{ATT}}_{\mathrm{DR}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^{n} \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[\Delta Y_i - \hat{\mu}_0(X_i) \right]}_{\text{OR residual}}$$

DR is consistent if *either* model is right — not both

$\hat{\mu}_0$ correct	$\hat{p}$ correct	Consistent?
$\checkmark$	$\checkmark$	$\checkmark$
$\checkmark$	$\times$	$\checkmark$
$\times$	$\checkmark$	$\checkmark$
$\times$	$\times$	**No**

"Doubly robust" $\neq$ doubly unbeatable.

Lab: ten estimators, one benchmark

`Labs/Lalonde-Covariates/lalonde_all_specs.do`

NSW treated $+$ CPS controls · Pre = 1975, Post = 1978

RCT benchmark: **$1,794**

Spec 0 (Naive) · Spec A (Additive X) · Spec B (Post$\times X$) · Spec C (Saturated)

OR by hand · OR `drdid` · IPW by hand · IPW `drdid` · DR by hand · DR `drdid`

A Monte Carlo to rank the estimators

Estimators:	TWFE, OR, IPW, DR
DGPs:	4 (vary whether OR model and pscore are correct)
n:	1,000 Replications: 10,000
Report:	Bias, RMSE, SE, coverage, CI length

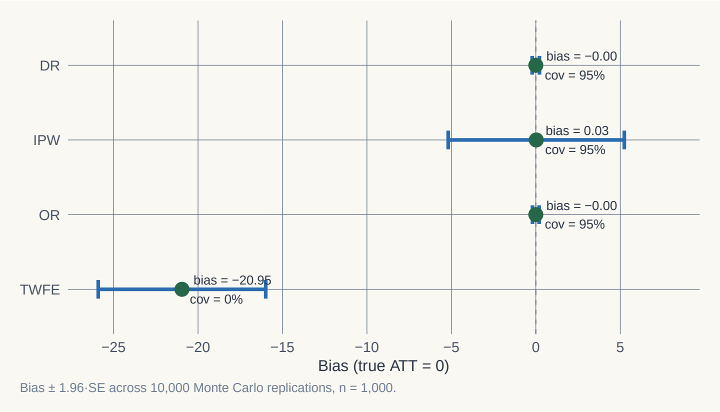
When does DR pay off? When does it break?

DGP 1: both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	-20.95	21.12	2.53	0.000	9.91
OR	-0.001	0.10	0.10	0.950	0.40
IPW	$+0.026$	2.77	2.66	0.952	10.44
DR	-0.001	0.11	0.11	0.947	0.41

TWFE: coverage zero. OR and DR: tight on truth. IPW: unbiased but wide.

DGP 1: the sampling distribution

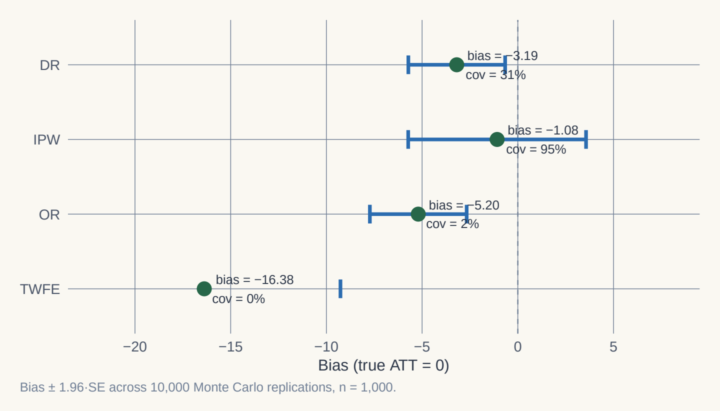


DGP 4: both models wrong

	Bias	RMSE	SE	Coverage	CI length
TWFE	-16.38	16.54	3.63	0.000	14.22
OR	-5.20	5.36	1.29	0.015	5.05
IPW	-1.08	2.66	2.37	0.949	9.31
DR	-3.19	3.45	1.29	0.308	5.07

DR undercovers at 0.31. IPW wins. Doubly robust $\neq$ doubly unbeatable.

DGP 4: the sampling distribution



What the Monte Carlo teaches

- **TWFE + additive X:** coverage zero in every DGP. **Stop using it as a default.**
- **OR:** efficient when right. Coverage 1.5% when wrong.
- **IPW:** less efficient, but robust to outcome misspecification.
- **DR:** best when one model is right. No better than IPW when both are wrong.

Adding X doesn't make your DiD credible

Adding X shifts the credibility question—

from **"does PT hold?"**

to **"is my model of $Y(0)$ or $p(X)$ correct?"**

You still have to answer it.

Now let's see it with real data

LaLonde (1986) · Dehejia & Wahba (2002)

A famous dataset where we **know the true effect.**

The benchmark: **$1,794**.

LaLonde (1986): the RCT found ≈\$800

- Job training program (National Supported Work Demonstration)
- Randomized — treated and control groups assigned by lottery
- RCT estimate: earnings effect ≈ **\$800**

A clean causal estimate. The benchmark exists.

LaLonde: dropped the RCT controls, used CPS instead

- Replaced the randomized controls with observational data (CPS, PSID)
- Re-ran every evaluator method from the literature
- Results ranged from $-$**\$16,000 to** $+$**\$3,000**

The true effect was \$800. Not one method recovered it.
"The available evidence calls into serious question the ability of nonexperimental evaluators…"

Empirical labor economics faces an evaluation problem.

Dehejia & Wahba (2002): propensity scores get close

- Need **two pre-treatment years** for all datasets — uses a shorter sample
- RCT benchmark on this shorter sample: **$1,794**
- Applied propensity score matching on the non-experimental sample
- Result: estimates close to $1,794

Selective use of the data + right method = nearly recovered the truth.

Our exercise: DiD with and without covariates

NSW treated + CPS controls · Pre = 1975 · Post = 1978

We know the effect: **$1,794**

Let's see which specs recover it — and which don't.

Naive TWFE: $3,621 — almost double the truth

```
reg re i.post##i.ever_treated, robust
```

Estimator	ATT	vs benchmark
Naive TWFE	**$3,621**	+$1,827

No covariates — the CPS controls trend very differently from NSW treated.

Additive X: still \$3,621

```
reg re i.post##i.ever_treated $X, robust
```

Estimator	ATT	vs benchmark
Additive X	**\$3,621**	$+$\$1,827

Adding X as time-invariant regressors does nothing here. The unit FE absorbs the level differences. The trend bias remains.

Post $\times X$: $1,711 — now in the right range

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.agecube ///
    i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree ///
    i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##i.u74, robust
```

Estimator	ATT	vs benchmark
Post $\times X$	**$1,711**	$-$$83

Saturated TWFE: $1,770

```
* First-difference, then regress dy on X + D x X interactions
reg dy $X ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.re74 ///
    i.ever_treated, robust
* Predict counterfactuals; ATT = mean(tau_sat) for treated
```

Estimator	ATT	vs benchmark
Saturated TWFE	**$1,770**	−$24

OR (HIT 1997): same answer as saturated TWFE

```
* Fit trend model on controls only
reg dy $X if ever_treated == 0, robust
predict dy_hat
* ATT = mean gap for treated
gen delta_hat = dy - dy_hat if ever_treated == 1
```

Estimator	ATT	vs benchmark
OR by hand / `drdid reg`	**\$1,770**	−\$24

Saturated TWFE and OR are algebraically equivalent in the 2-period case.

IPW (Abadie 2005): $1,861

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

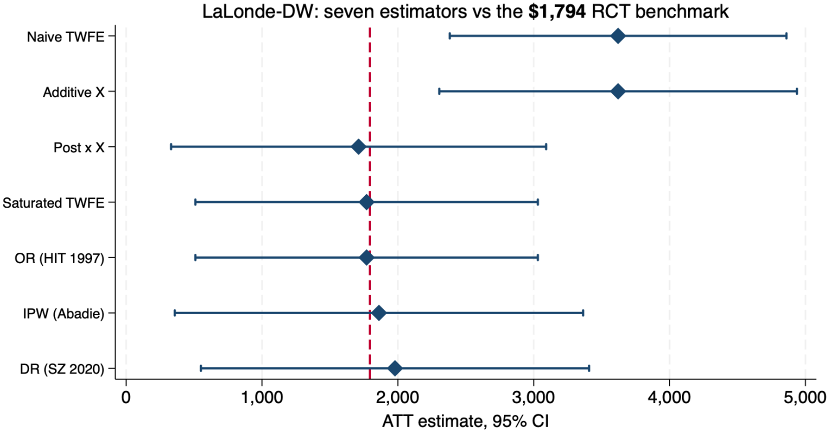
Estimator	ATT	vs benchmark
IPW	**$1,861**	+$67

DR (Sant'Anna–Zhao 2020): $1,979

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Estimator	ATT	vs benchmark
DR	**$1,979**	+$185

All seven estimators vs the $1,794 benchmark



Red dashed line = RCT benchmark ($1,794). Saturated TWFE uses OR's drdid SE.
by-hand and drdid estimates agree to the dollar for OR and IPW.

Covariates are not robustness checks — they are design

- People say: *"if adding covariates changes results, be suspicious."*
- That gets it backwards.
- We **know** the effect is $1,794.
- Without covariates, or with additive covariates, we get $3,621 — *wrong*.
- The correct specs recover $1,770–$1,979 — *right*.

Covariates fix broken parallel trends. Changing your result is not a bug. It is the design working.

The question is never

“did I control for X?”

The question is always

“did I get the model right?”